

Set 15: Product Sales Analysis

Department	AIDS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
Name	K. Deeksha Sree
Register Number	192424097
Date	08 October 2026
Day	Thursday

Aim

To analyse monthly product sales using a grouped bar chart, a stacked area chart, and a table, and to build an interactive dashboard that combines all three.

Dataset: Monthly Product Sales

Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	1100	2000	2200
2	Product B	1500	2400	1800
3	Product C	1200	1600	1400

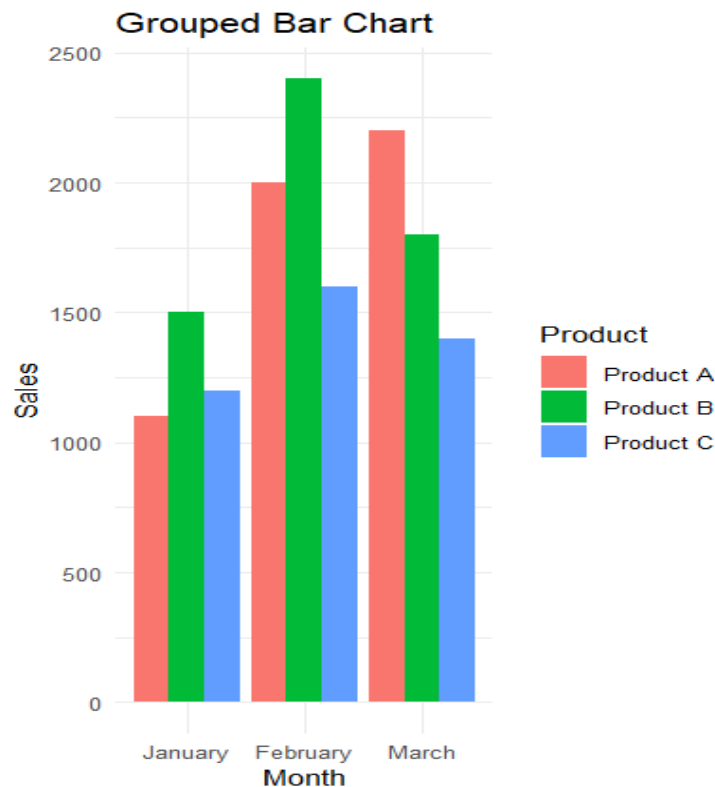
Note: The dataset text was jumbled when copied, so Product A's January sales were read as 1100. Please check this against your question paper.

PROGRAM 1: Grouped Bar Chart of Product Sales

Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.

```
# Dataset
sales <- data.frame(
  Product = rep(c("Product A", "Product B", "Product C"), each = 3),
  Month   = factor(rep(c("January", "February", "March"), times = 3),
                  levels = c("January", "February", "March")),
  Sales   = c(1100, 2000, 2200,
              1500, 2400, 1800,
              1200, 1600, 1400)
)

# Grouped bar chart
library(ggplot2)
ggplot(sales, aes(x = Month, y = Sales, fill = Product)) +
  geom_bar(stat = "identity", position = "dodge") +
  geom_text(aes(label = Sales), position = position_dodge(0.9), vjust = -0.3) +
  labs(title = "Product Sales in the First Quarter",
       x = "Month", y = "Sales", fill = "Product") +
  theme_minimal()
```



Output 1: Grouped Bar Chart of First Quarter Product Sales

Tableau steps: unpivot January, February and March Sales into Month and Sales (Pivot). Drag Month to Columns, Sales to Rows, and Product Name to Color. Then drag Product Name to Columns next to Month to group the bars, show Sales as labels, and title the sheet "Product Sales in the First Quarter".

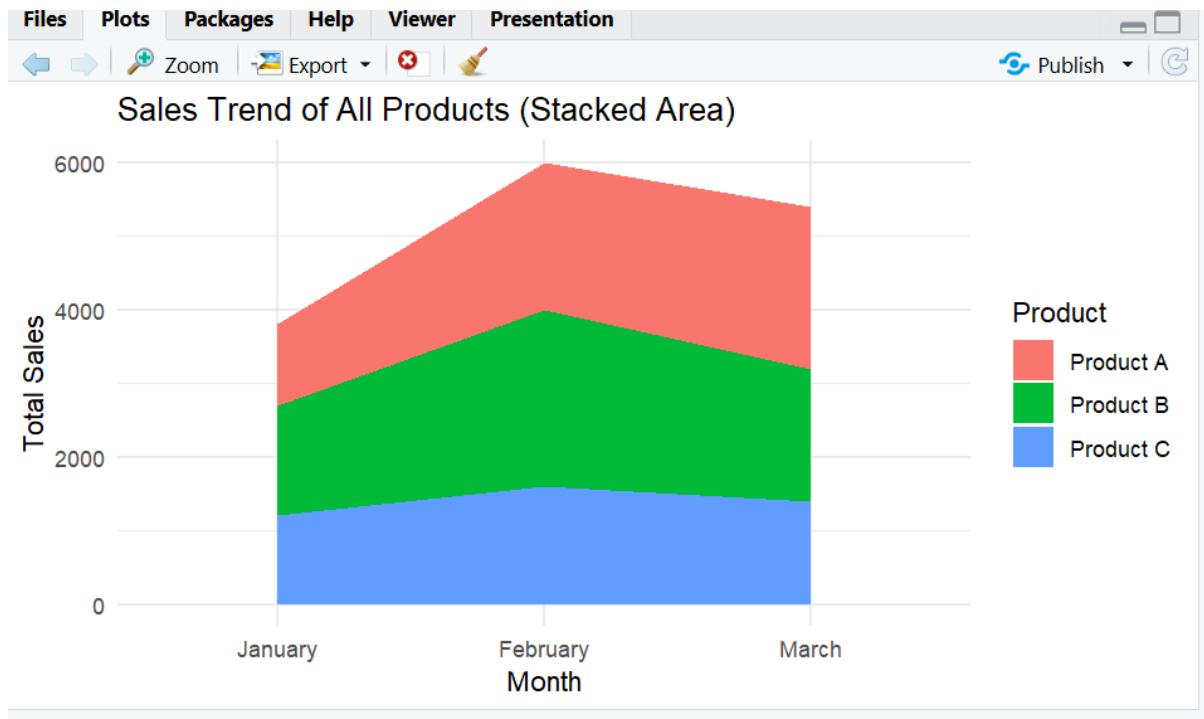
Observation: Product A's sales rise every month (1100 to 2200). Product B has the highest sales in February (2400) but drops in March (1800). Product C has the lowest sales in every month except January, where it is above Product A.

PROGRAM 2: Stacked Area Chart of Sales Trend

Generate a stacked area chart to represent the overall sales trend for all products over the three months.

```
# Dataset
sales <- data.frame(
  Product = rep(c("Product A", "Product B", "Product C"), each = 3),
  Month   = factor(rep(c("January", "February", "March"), times = 3),
    levels = c("January", "February", "March")),
  Sales   = c(1100, 2000, 2200,
    1500, 2400, 1800,
    1200, 1600, 1400)
)

# Stacked area chart
library(ggplot2)
ggplot(sales, aes(x = Month, y = Sales, fill = Product, group = Product)) +
  geom_area() +
  labs(title = "Sales Trend of All Products (Stacked Area)",
    x = "Month", y = "Total Sales", fill = "Product") +
  theme_minimal()
```



Output 2: Stacked Area Chart of Overall Sales Trend

Tableau steps: drag Month to Columns and Sales to Rows, drag Product Name to Color, then change the mark type to Area. Tableau stacks the areas by default.

Observation: Total sales rise from 3,800 in January to 6,000 in February, then fall to 5,400 in March. February is the peak month. The March drop comes from Products B and C, while Product A keeps growing.

PROGRAM 3: Table of Monthly Sales Data

Build a table to show the monthly sales data for each product. Label the table elements.

```
# Table of monthly product sales
sales_table <- data.frame(
  "Product ID"      = 1:3,
  "Product Name"    = c("Product A", "Product B", "Product C"),
  "January Sales"   = c(1100, 1500, 1200),
  "February Sales"  = c(2000, 2400, 1600),
  "March Sales"     = c(2200, 1800, 1400),
  check.names = FALSE
)
print(sales_table)
View(sales_table)
```

	Product ID	Product Name	January Sales
1	1	Product A	1100
2	2	Product B	1500
3	3	Product C	1200

Showing 1 to 3 of 3 entries, 5 total columns

Summary for January Sales: [1, 3], 0% NA

Output 3: Monthly Product Sales Table

Tableau steps: create a new sheet, drag Product Name to Rows and Month to Columns, and drag Sales to Text. Turn on Analysis > Totals > Show Column Grand Totals to display the monthly totals.

Observation: Product B has the highest quarter total (5,700), followed by Product A (5,300) and Product C (4,200). The all-product total for the quarter is 15,200.

PROGRAM 4: Interactive Sales Dashboard

Develop a dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data.

A. Dashboard in R (Shiny)

```
library(shiny)
library(ggplot2)

sales <- data.frame(
  Product = rep(c("Product A", "Product B", "Product C"), each = 3),
  Month = factor(rep(c("January", "February", "March"), times = 3),
    levels = c("January", "February", "March")),
  Sales = c(1100, 2000, 2200, 1500, 2400, 1800, 1200, 1600, 1400)
)

ui <- fluidPage(
  titlePanel("Product Sales Dashboard"),
  sidebarLayout(
    sidebarPanel(
      checkboxGroupInput("prod", "Select Products:",
        choices = unique(sales$Product),
        selected = unique(sales$Product))
    ),
    mainPanel(
      fluidRow(
        column(6, plotOutput("barPlot")),
        column(6, plotOutput("areaPlot"))
      )
    )
  )
)
```

```

    ),
    h4("Monthly Sales Table"),
    tableOutput("salesTable")
  )
)
)

server <- function(input, output) {
  filtered <- reactive({
    req(input$prod)
    sales[sales$Product %in% input$prod, ]
  })

  output$barPlot <- renderPlot({
    ggplot(filtered(), aes(x = Month, y = Sales, fill = Product)) +
      geom_bar(stat = "identity", position = "dodge") +
      labs(title = "Grouped Bar Chart", x = "Month", y = "Sales") +
      theme_minimal()
  })

  output$areaPlot <- renderPlot({
    ggplot(filtered(), aes(x = Month, y = Sales, fill = Product, group = Product)) +
      geom_area() +
      labs(title = "Stacked Area Chart", x = "Month", y = "Total Sales") +
      theme_minimal()
  })

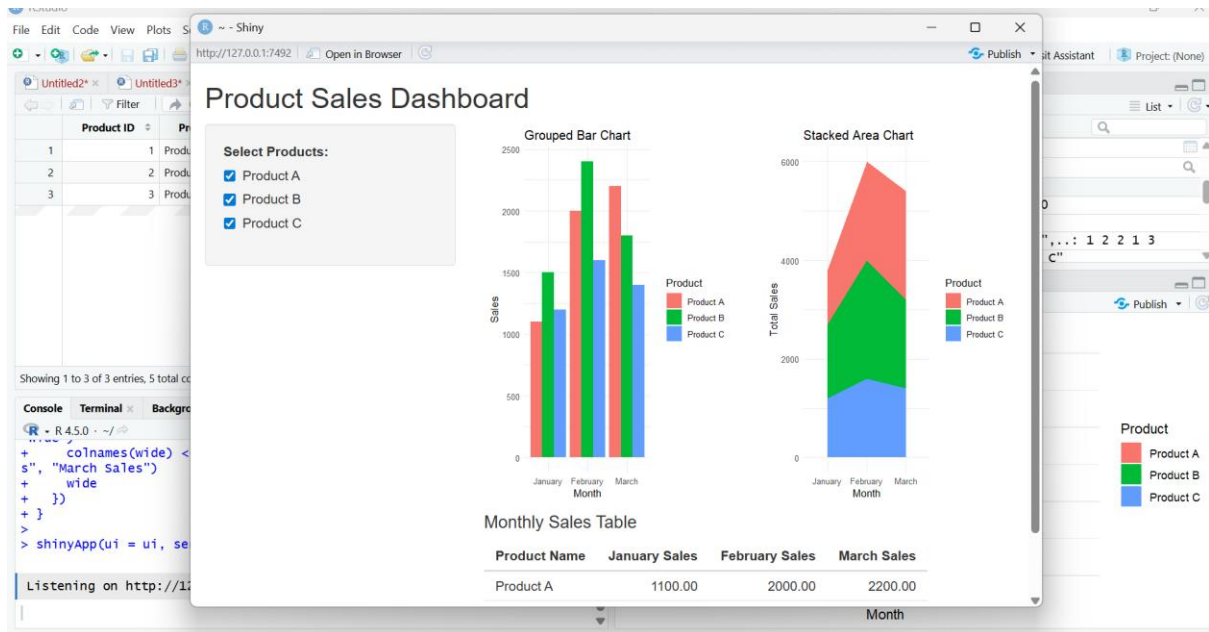
  output$salesTable <- renderTable({
    d <- filtered()
    wide <- reshape(d, idvar = "Product", timevar = "Month", direction = "wide")
    colnames(wide) <- c("Product Name", "January Sales", "February Sales", "March
Sales")
    wide
  })
}

shinyApp(ui = ui, server = server)

```

B. Dashboard in Tableau

1. Open Tableau and connect to the Excel/CSV file, then pivot the three month columns into Month and Sales.
2. Create three worksheets: Grouped Bar (Program 1), Stacked Area (Program 2), and Table (Program 3).
3. Click New Dashboard and set the size to Automatic (or Fixed 1000 x 800).
4. Drag the Grouped Bar and the Stacked Area side by side at the top and the Table across the bottom.
5. For each sheet, click the dropdown arrow and select Use as Filter, so clicking a product or month filters the other views.
6. Add a Product Name quick filter (Show Filter) and apply it to all worksheets.
7. Add the dashboard title "Product Sales Dashboard" and tidy the legends and layout.
8. Save the workbook and use Presentation Mode to test the interactions.



Output 4: Interactive Product Sales Dashboard

Observation: The dashboard combines the grouped bar chart, stacked area chart, and table in one view. Selecting products updates all three views, so monthly sales and trends can be compared together.

Procedure

1. Create the product sales data frame in R with Product, Month and Sales.
2. Use ggplot2 with `geom_bar(position = "dodge")` to create the grouped bar chart.
3. Use `geom_area()` to create the stacked area chart of the sales trend.
4. Create a data frame/table to display the monthly sales of each product.
5. Build the Shiny dashboard (or the Tableau dashboard) with product filters to explore the sales data interactively.

Result

The monthly product sales were successfully analysed using a grouped bar chart, a stacked area chart, and a table. An interactive dashboard was created by combining all three views for exploring the sales data.